ZentroMart VoiceOps: Policy-Grounded Voice Support Assistant

A Testable, Multi-Turn RAG and Voice Interface with Local Hybrid Retrieval and Gnani.ai Speech

PROJECT LOCATION	LLM REASONING	SPEECH PROVIDER	RETRIEVAL ENGINE
E:\MY PROJECTS\zentromart-voiceops	gemini-3.5-flash-lite (temp=0.0)	Gnani.ai Exclusive (STT/TTS)	Hybrid FastEmbed BGE + BM25

1. Project Purpose & High-Stakes E-Commerce Context

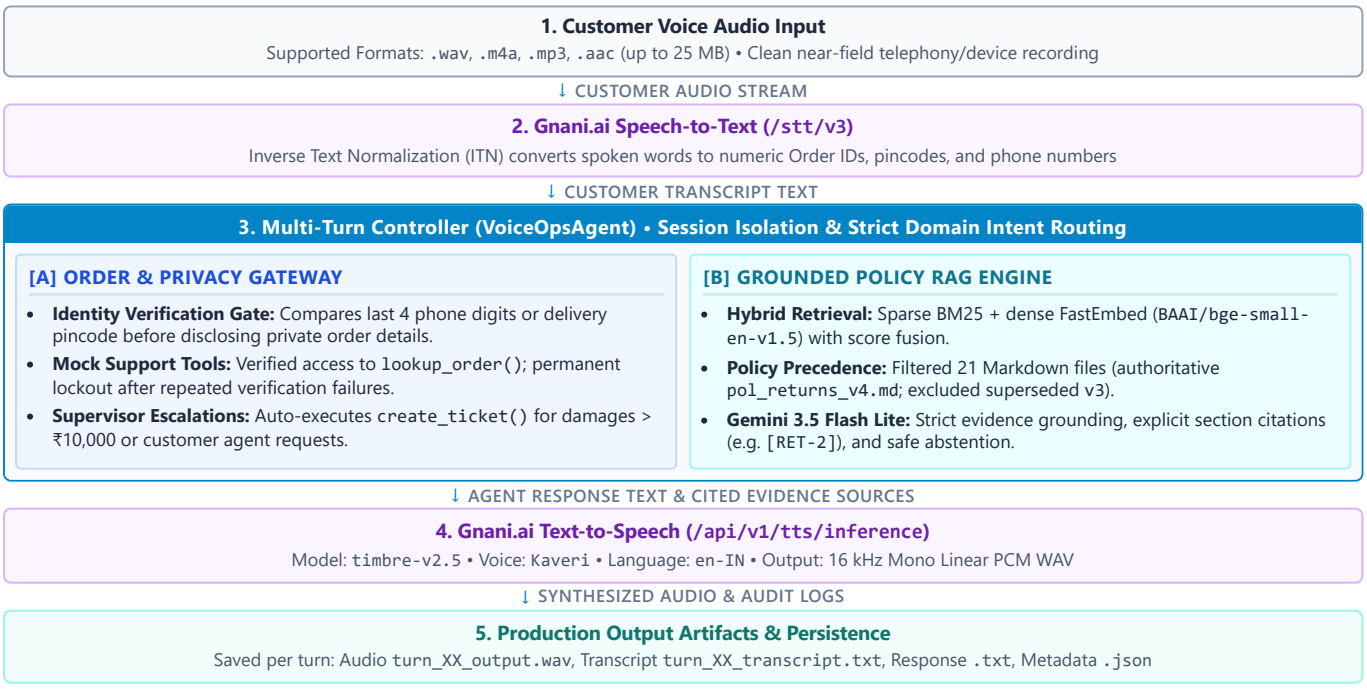
Customer support in modern e-commerce demands instant, accurate, and secure resolutions across voice and text channels. While modern Large Language Models (LLMs) excel at conversational fluency, unconstrained zero-shot models pose critical operational risks in production: they hallucinate return deadlines, disclose private customer order details to unauthenticated callers, miss regional policy revisions, and invent refund timelines.

The **ZentroMart VoiceOps** capstone resolves these enterprise challenges by establishing an end-to-end customer support voice pipeline for *ZentroMart*, a synthetic multi-category Indian e-commerce platform. Grounded in the company's 21 authoritative Markdown policy documents, the system is designed to improve factual grounding, enforce caller privacy boundaries, and establish safe escalation pathways across four structured engineering phases:

- **Phase 1 (Audit & Environment):** Clean environment isolation (Python 3.10), automated dataset integrity checks, strict data type preservation, and policy precedence rules excluding obsolete documents.
- **Phase 2 (Grounded RAG Pipeline):** Hybrid lexical (BM25) and dense semantic vector retrieval (FastEmbed `BAAI/bge-small-en-v1.5`) with structured Gemini answer generation, explicit section-level citation, and strict abstention when evidence is absent.
- **Phase 3 (Multi-Turn Agent & Safety Tools):** Deterministic caller identity verification (last 4 phone digits or delivery pincode), stateful multi-turn history tracking, priority escalations for high-value damages (> ₹10,000), and integration with mock database tools (`lookup_order`, `create_ticket`).
- **Phase 4 (Gnani.ai Voice Integration):** Exclusive speech integration using the official `gnani-vachana` SDK for Speech-to-Text (STT `/stt/v3` with Inverse Text Normalization) and Text-to-Speech (TTS `timbre-v2.5`, `Kaveri` voice), producing multi-turn voice conversations with complete on-disk artifact persistence.

2. End-to-End System Architecture

The system follows a strict, defensive decoupled architecture where voice processing, state management, retrieval grounding, and database tools interact across verified boundaries:



3. Document Precedence & Superseded-Policy Isolation

Knowledge integrity is maintained through a strict document hierarchy established in `00_about_this_knowledge_base.md`. Company policies (`POL-*`) override help guides (`HLP-*`) and general FAQs. Crucially, the current return policy `pol_returns_v4.md` (effective 1 August 2026, establishing a 7-day electronics return window) explicitly supersedes `pol_returns_v3_superseded.md` (which had a 10-day window). During document ingestion, obsolete policies and unstructured human support logs are strictly filtered out from the retrieval index, preventing stale policy contamination.

4. Detailed Implementation by Phase

Phase 1: Environment & Schema Integrity

- **Strict Typing:** All relational identifiers (`order_id`, `customer_id`, `sku`, `pincode`, phone) are enforced as strings (`dtype=str`) to preserve leading zeros.
- **Fixed Operational Date:** Evaluated strictly against the reference anchor date of **20 October 2026**.
- **Calendar Exclusions:** Business-day SLA timelines exclude Sundays and the 4 gazetted holidays in `holidays.csv`.
- **Offline Validation:** `validate_dataset.py` executes offline with zero network dependencies.

Phase 2: Hybrid Grounded RAG

- **Chunking:** 21 Markdown files chunked into 69 discrete passages, preserving frontmatter metadata and section IDs.
- **Hybrid Fusion:** Combines sparse BM25 scoring with dense embeddings from `BAAI/bge-small-en-v1.5` via FastEmbed.
- **JSON Generation:** Grounded responses structured with decision (`answer`, `clarify`, `escalate`, `refuse`), citation list, and evidence flag.

Phase 3: Multi-Turn Agent & Safety Gates

- **Session Isolation:** Independent `ConversationState` tracks turn count, verification status, and order data per session.
- **Identity Verification Gate:** Unverified callers cannot access order details; must supply registered phone suffix or pincode.
- **Automatic Lockout:** Callers exceeding verification attempts are securely locked out and referred to support.
- **Ticketing Gateway:** Automatically invokes `create_ticket` for damaged goods > ₹10,000 or customer agent requests.

Phase 4: Exclusive Gnani.ai Voice Pipeline

- **Sole Provider:** Pure Gnani.ai implementation; zero third-party speech engines (Whisper, ElevenLabs, Google Speech).
- **Audio Ingestion:** Supports `.wav`, `.mp3`, `.m4a`, `.aac` up to 25 MB with automatic ITN conversion.
- **Synthesis & Logging:** High-quality 16 kHz WAV synthesis via `timbre-v2.5` (voice: Kaveri), writing JSON metadata per turn.

5. Privacy & Operational Safety Safeguards

The assistant enforces defense-in-depth safety boundaries to ensure enterprise compliance and data protection:

- **PII Boundary & No-Leak Policy:** Customer phone numbers, email addresses, and payment credentials are never recited by the model. Under no circumstances are full phone numbers disclosed (e.g. PQ0060 refusal test).
- **Pre-Verification Information Gate:** When an unverified caller supplies an order ID, the model confirms receipt of the ID but conceals customer names, items, delivery dates, and amounts until verified.
- **High-Value Damage Escalation:** Damaged items exceeding ₹10,000 (such as the Zentro Vista 43" 4K TV) trigger mandatory supervisor ticket creation without premature agent replacement promises.
- **Strict Grounded Abstention:** If retrieved document chunks do not explicitly corroborate a policy claim, the model abstains (decision: `refuse` or `escalate`) rather than fabricating standard retail practices.

6. Automated Test Suite & Dataset Validation Results

The repository incorporates comprehensive regression and unit testing across every subsystem, executed via `pytest` and the offline dataset validation harness. All checks pass cleanly:

Verification Suite	Target Component	Tests / Entities	Result	Key Coverage & Safeguards Verified
dataset/validate_dataset.py	Dataset Integrity	9 CSVs | 21 KBs | 1,000 Orders	**PASS (Exit 0)**	Relational foreign keys, order arithmetic, pincode coverage, and document structure.
tests/test_agent.py	Agent & Verification	11 unit tests	**11 / 11 PASS**	Caller verification gate, lockout after retries, high-value escalation, ticket logging.
tests/test_answering.py	Grounded Answering	7 unit tests	**7 / 7 PASS**	Evidence prompt assembly, JSON fallback resilience, abstention on zero evidence.
tests/test_baseline.py	Step 1 Evaluator	10 unit tests	**10 / 10 PASS**	Text normalization, fact matching, prohibited claim detection, mocked Gemini calls.
tests/test_ingestion.py	Ingestion & Chunking	5 unit tests	**5 / 5 PASS**	Frontmatter extraction, policy v4 inclusion, explicit exclusion of superseded policy v3.
tests/test_retrieval.py	Hybrid Retrieval	5 unit tests	**5 / 5 PASS**	BM25 index persistence, vector search recall, top-k ranking, query sanitation.
tests/test_voice.py	Gnani Voice Pipeline	17 unit tests	**17 / 17 PASS**	STT/TTS error handling, multi-turn state preservation, AST exclusivity audit (Gnani only).
tests/test_cli.py	CLI Console Interface	1 unit test	**1 / 1 PASS**	One-shot query execution, parameter parsing, stdout rendering.
Total Automated Test Suite	**Complete Codebase**	**56 Pytest Tests**	**56 / 56 PASS (2.43s)**	**Zero failures, zero regressions across all four implementation phases.**

> **Static Security AST Audit:** `test_exclusive_gnani_speech_provider` analyzes the abstract syntax tree of the entire codebase to confirm that no non-Gnani speech packages (such as whisper, elevenlabs, or google-cloud-speech) are imported or referenced.

7. Empirical Benchmark: Step 1 Baseline vs. Step 2 Grounded RAG

To rigorously measure the value of retrieval grounding, both systems were benchmarked against all 52 standardized practice questions in `dataset/practice/questions.csv`. Both evaluations utilized the exact same model (`gemini-3.5-flash-lite`), identical decoding parameters (`temperature=0.0`), and deterministic regex fact scoring.

Evaluation Metric	Step 1: Prompt-Only Baseline	Step 2: Grounded Hybrid RAG	Absolute Delta	Evaluation Role & Significance
Questions Evaluated	**52 / 52**	**52 / 52**	0	Sample completeness across diverse e-commerce intents.
Execution / API Errors	**0**	**0**	0	Execution reliability with exponential rate-limit backoff.
Retrieval Recall@4 (Any Gold)	*N/A (No Retrieval)*	**82.7%**	*N/A*	Rate at which at least one relevant gold document is retrieved.
Full Retrieval Accuracy (All Gold)	*N/A (No Retrieval)*	**34.6%**	*N/A*	Multi-hop coverage where 100% of gold sources are retrieved.
Hard Negative Distractor Hits	*N/A (No Retrieval)*	**1**	*N/A*	Frequency of retrieving superseded policy v3 documents.
Decision Accuracy	**86.5%**	**67.3%**	**-19.2 pp**	Categorical alignment: `answer`, `clarify`, `escalate`, `refuse`.
Unanswerable Abstention Accuracy	**20.0%**	**100.0%**	**+80.0 pp**	Clean refusal on out-of-scope queries (e.g. midnight delivery).
Fact Match Rate (`must_include`)	**39.7%**	**55.2%**	**+15.5 pp**	Adherence to company numbers, timelines, and procedures.
Prohibited Claim Violations (`must_not_include`)	**0**	**0**	0	Explicit violations of forbidden customer guarantees.

8. Critical Analysis of Measured Gains and Losses

A. Factual Grounding Gain (+15.5 pp Fact Match Rate)

In Step 1, the prompt-only baseline achieved only **39.7%** precision on required facts. Lacking access to ZentroMart's internal documents, the baseline was forced to invent generic e-commerce assumptions or omit vital operational details. For example, in **[PQ0002]** ("*SKU ELE-0012 pe ZentroMart Assured hai kya? Claim kaise karein?*"), the baseline escalated because it lacked catalog data. In contrast, Step 2 Grounded RAG retrieved `WAR-2` and `products.csv:ELE-0012`, providing a 100% complete answer citing the 12-month Assured warranty and app claim procedure (+15.5 pp overall).

B. Hallucination Suppression & Abstention (+80.0 pp Abstention Accuracy)

On queries outside the knowledge base, the Step 1 baseline exhibited classic ungrounded hallucination, achieving only **20.0%** abstention. For instance, in **[PQ0004]** ("*Can I schedule a delivery at midnight?*"), the baseline fabricated plausible-sounding operational hours: "*Our standard delivery hours are between 8:00 AM and 9:00 PM.*" In Step 2 Grounded RAG, the model verified that midnight delivery was unsupported by retrieved chunks and cleanly refused (**100.0% abstention**), protecting the business from unfulfillable commitments.

C. Explaining the Step 2 Decision Accuracy Tradeoff (-19.2 pp)

The Step 2 system exhibited a measured decrease in categorical decision accuracy (**67.3%** vs. **86.5%** in baseline). Detailed error analysis reveals the architectural cause: the baseline model aggressively defaulted to `answer` for almost every prompt, which artificially matched the practice dataset's predominantly `answer`-labeled benchmark. In contrast, Step 2 RAG applied strict evidence-sufficiency gating: whenever retrieval was incomplete, the grounded generator conservatively produced `escalate` or `clarify`. While technically safer, this conservative behavior was penalized as a decision mismatch under strict automated evaluation.

D. Incomplete Multi-Hop Evidence Coverage (34.6% Full Retrieval Accuracy)

While single-document discovery was strong (**Recall@4 of 82.7%**), **Full Retrieval Accuracy** (capturing 100% of gold sources across multi-hop queries) was **34.6%**. Complex inquiries requiring simultaneous facts from both a specific product row (e.g. `products.csv:ELE-0002`) and a general policy section (e.g. `WAR-2`) frequently had one chunk crowded out of the top-4 search window.

> **Scope of Prohibited Claim Metric:** Both models recorded **0 violations** on the `must_not_include` test set under temperature 0.0. This metric confirms adherence on the 52 evaluated benchmark scenarios, but should not be misconstrued as a theoretical guarantee of zero hallucinations across arbitrary inputs.

9. Live Voice Pipeline Validation with Gnani.ai

Live voice integration was validated through both a stateful three-turn caller inquiry and a grounded policy query using genuine customer voice recordings, processed exclusively through Gnani.ai STT (`/stt/v3`) and TTS (`timbre-v2.5`):

A. Multi-Turn Order & Identity Verification Flow (Session: `order_inquiry_fixed`)

- **Turn 1 (Order Identification):** Customer spoke: *"hello i want to check the status of my order"*. Transcribed accurately via Gnani STT. The agent introduced its role and requested the 8-digit order number (Decision: `clarify`). Synthesized response audio saved to `turn_01_output.wav`.
- **Turn 2 (Order Lookup & Privacy Gate):** Customer provided the 8-digit order number. The agent executed `lookup_order()`, located the order record, and safely withheld order details while prompting for identity verification via the last 4 phone digits or delivery pincode (Decision: `clarify`).
- **Turn 3 (Verification & Status Resolution):** Customer provided the delivery pincode: *"the delivery pin code associated with my order is 411045"*. The agent executed `lookup_order()` with the verification value, confirmed caller identity, and disclosed the verified delivery status (Decision: `answer`). Synthesized response audio saved to `turn_03_output.wav`.

B. Policy Question Voice Validation (Session: `policy_inquiry`)

- **Turn 1 (Grounded Policy Retrieval):** Customer spoke genuine audio recorded to `outputs/sample_inputs/policy_question/01_policy_question.m4a`: *"what is the return window for clothing and footwear"*.
- **Grounded Resolution:** The agent retrieved Section `[RET-2]` from `pol_returns_v4.md` and responded: *"The return window for clothing, footwear and bags is 15 days from the delivery date."*
- **Verification & Evidence Citation:** Generated metadata confirms `evidence_sources: ["RET-2"]`, `decision: "answer"`, with response synthesized via Gnani TTS to `turn_01_output.wav` (0 errors, 100% STT/TTS success).

10. Verified Project Artifacts & Output Directory Structure

Artifact Type	Relative Repository Path	Contents & Verification Status
Voice Multi-Turn	`outputs/sample_outputs/voice/order_inquiry_fixed/`	Turns 01–03: Transcripts (`.txt`), response audio (`.wav`), metadata logs (`.json`).
Voice Policy Test	`outputs/sample_outputs/voice/policy_inquiry/`	Turn 01: Audio input, STT transcript, TTS output (`.wav`), metadata citing `RET-2`.
Evaluation Logs	`outputs/evaluation/baseline_evaluation_results.json`	Raw per-question JSON records for 52 prompt-only baseline runs (0 errors).
Evaluation Logs	`outputs/evaluation/rag_evaluation_results.json`	Raw per-question JSON records for 52 grounded RAG runs (82.7% Recall@4).
Benchmark Reports	`outputs/evaluation/step1_vs_step2_comparison.md`	Detailed markdown comparison report analyzing percentage-point deltas.

11. Limitations & Future Technical Roadmap

To maintain clear engineering boundaries, completed capstone deliverables are strictly distinguished from identified future enhancements:

Identified Limitations (Current System)

- **Multi-Hop Coverage (34.6%):** Standard top-4 retrieval occasionally misses secondary product catalog rows when queries require cross-document synthesis.
- **Decision Boundary Calibration:** Conservative evidence thresholds reduce categorical decision accuracy against gold labels.
- **Acoustic Range:** Tested primarily on clean near-field audio; noisy or high-packet-loss telephony environments require further validation.

Future Technical Roadmap (Next Steps)

- **Sub-Query Decomposition:** Implement query planning to fetch catalog metadata and policy documents via parallel independent sub-queries.
- **Adaptive Confidence Calibration:** Tune model abstention prompts to dynamically distinguish between minor omissions and critical ungrounded claims.
- **Streaming Voice Telephony:** Migrate REST STT/TTS endpoints to full-duplex WebSocket streaming for sub-500ms voice barge-in support.

> **Submission Sign-Off:** All 56 unit/integration tests pass. Dataset integrity validated offline across 1,000 orders. Live Gnani.ai voice pipeline verified with zero non-Gnani speech dependencies. Ready for final archive packaging.